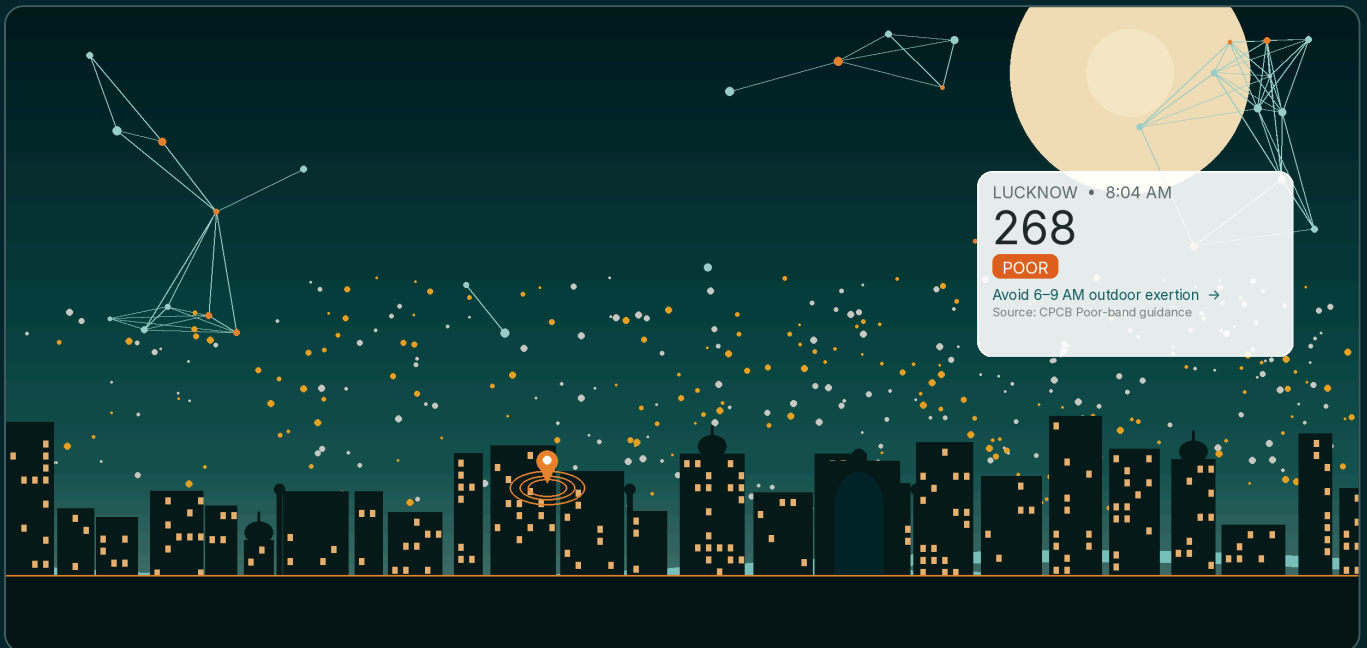




VayuSathi

AI for Cleaner Air. Smarter Decisions. Healthier Cities.

AI-Powered Air Quality Monitoring & Health Advisory Assistant
Starting with Lucknow, India



◆ **SDG 11 • 3 • 13**

Sustainable cities • Health • Climate

◆ **PROTOTYPE**

5 workflows • RAG • 9 UI screens

◆ **HINDI + ENGLISH**

Voice-ready • Low-text cards

Mandeep Singh

University of Lucknow • Lucknow, Uttar Pradesh

1M1B x IBM SKILLSBUILD x AICTE
AI FOR SUSTAINABILITY VIRTUAL INTERNSHIP
FINAL PROJECT • JULY-SEPTEMBER 2026



INSIDE THIS REPORT

Contents

A visual story in four acts — **the air problem** → **the human pain** → **how the AI works** → **the impact it can create**. Every topic begins on a fresh page.

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→	Closing — from AQI numbers to everyday action	30

Evidence badges used in this report:

DEMO DATA

PROJECTED TARGET

CONCEPTUAL UI

PROPOSED ARCHITECTURE

FUTURE SCOPE

Declaration — Mandeep Singh, University of Lucknow



§1 • EXECUTIVE SUMMARY

VayuSathi at a Glance

Lucknow's winter air regularly turns **Poor-Very Poor**. People see an AQI number — VayuSathi tells them **what to do about it**: when to go out, how to commute, where to exercise, in **Hindi or English**.

PROBLEM

AQI data exists. Actionable guidance doesn't.

"AQI 268" confuses more than it helps — no personalisation, no Hindi, no plan.

SOLUTION

AI converts AQI + context into personal action.

Understand → Personalise → Plan → Act, in ~30 seconds.

USERS

Students • families • workers • athletes • schools

Six personas, one assistant — including Hindi-first outdoor workers.

AI

LLM + RAG + prompt engineering

IBM Granite-class models, grounded in CPCB / WHO / IMD guidance with citations.

LANGUAGE

English + Hindi, Hinglish-friendly

Every answer ships with a Hindi summary. Voice-ready, low-text cards.

IMPACT

Safer routines + cleaner mobility

Fewer high-exposure hours, smarter commutes, informed communities.

PROJECTED TARGET

The story this report tells

§2-§4

The air problem

Lucknow's pollution, gaps & data

§5-§6

The human pain

Personas, journey & SDGs

§7-§15

How the AI works

Features, architecture, RAG, UI

§16-§17

How it stays safe

Honesty & trust layer

§18-§21

Impact & future

Outcomes, roadmap, scale

Project maturity

✓ CONCEPT

Defined & designed

✓ PROTOTYPE

Prompts + UI + arch

NEXT • PILOT

Campus + school

FUTURE • SCALE

City playbook



§2 • THE CORE NARRATIVE

From a scary number to a clear plan

The entire project in one storyboard: what happens **today**, what is **missing**, and how **VayuSathi** closes the gap.

TODAY

AQI 268 **POOR**

User sees a number in an app —
in English, with a colour dot.

**"What should I actually
do?"**

Walk? Mask? Metro?
Reschedule? Nobody says.

THE GAP

No:

- ✗ personalisation by profile
- ✗ local-language guidance
- ✗ forward (48-hr) planning
- ✗ actionable commute advice
- ✗ school / campus protocol

Result: the most exposed people —
riders, vendors, children, athletes —
improvise.

VAYUSATHI

1 Understand
AQI 268 = Poor: what it
means for breathing

2 Personalise
Risk for *you* — asthma,
child, rider, athlete...

3 Plan
Best windows, commute &
indoor swaps

4 Act
Hindi summary + low-
carbon tip + alert

Design principle — never colour alone

Every AQI visual in this report carries **number + category + plain-language meaning**. The spectrum below recurs throughout the document as a shared visual language.





§3 • PROBLEM NARRATIVE

The Lucknow Air Challenge

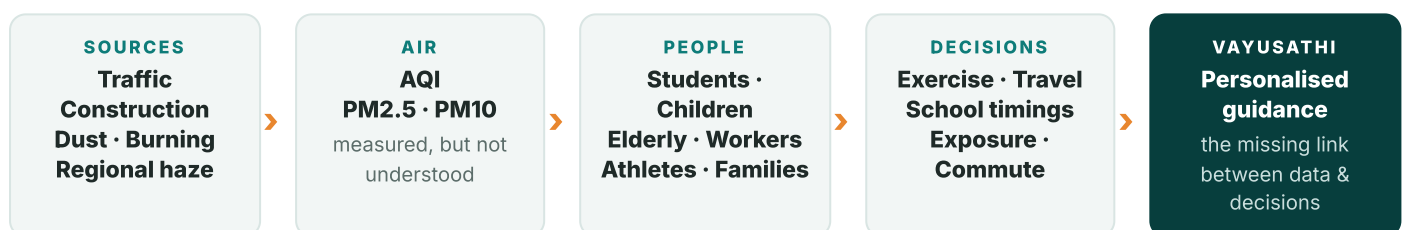
HOW MIGHT WE...?

...use AI to translate complex, fast-changing air-quality data into clear, personal, local-language health and mobility guidance — so that students, families, workers and schools in Lucknow can reduce harmful exposure and live more sustainably?

What drives poor air — and what it disrupts



Pollution ecosystem — why guidance must connect everything



Scope boundary: VayuSathi builds no sensors and replaces no official forecast — it is a **decision-support and awareness layer** over CPCB / IMD public data, giving general preventive information only (never diagnosis).



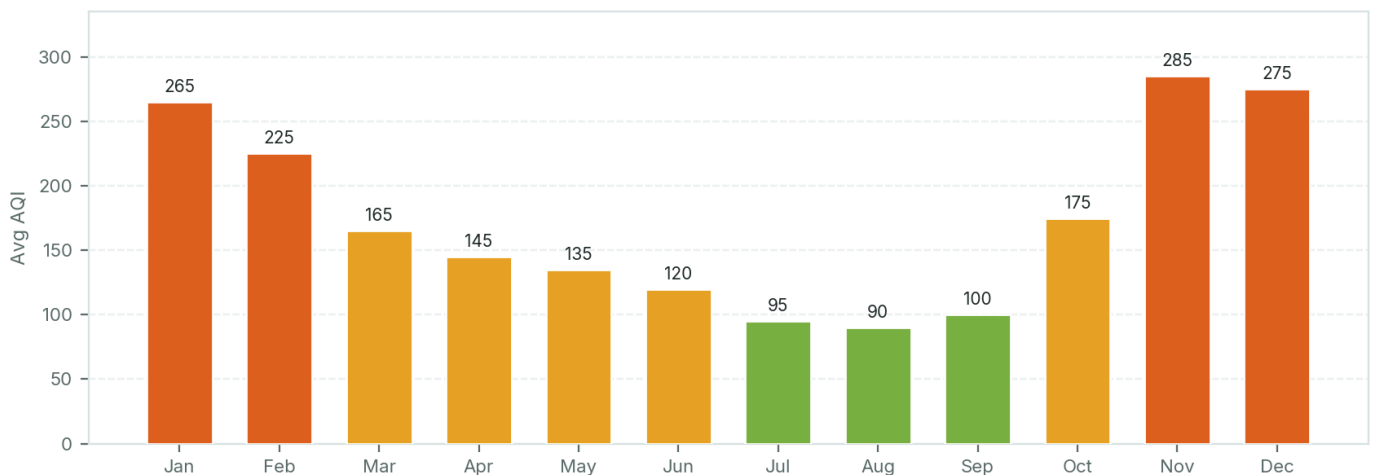
§ 4 • EVIDENCE, HONESTLY LABELLED

Air Quality in Numbers

Illustrative seasonal patterns for Lucknow (inspired by typical CPCB trends) — showing **when** guidance matters most. **DEMO DATA** All values on this page are illustrative — not live monitoring.

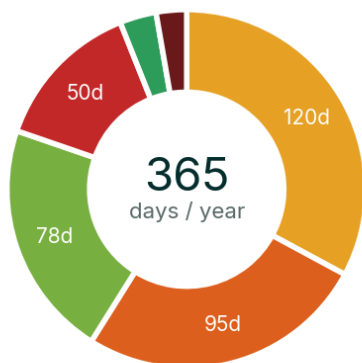
Lucknow — seasonal AQI pattern

Illustrative demo data · winter peak Oct–Feb needs daily guidance



Seasonal AQI — October–February forms a sustained high-risk window where daily guidance matters most.

Days per AQI category

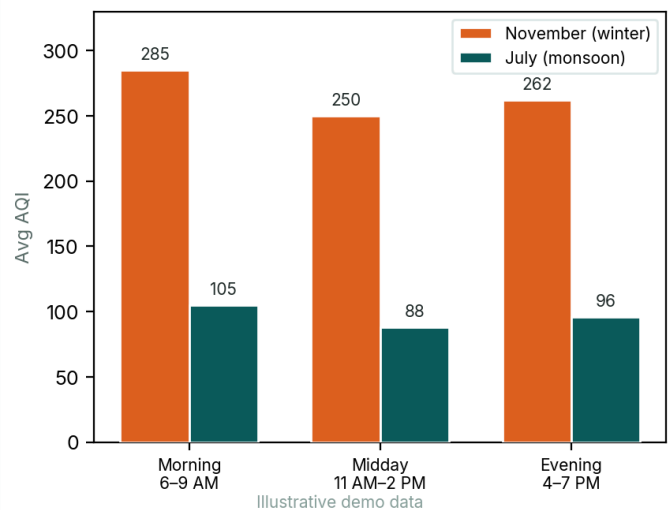


Moderate 101–200 Satisfactory 51–100 Good 0–50
Poor 201–300 Very Poor 301–400 Severe 401+

Illustrative demo data

~215 of 365 days sit above “Satisfactory” — guidance is a year-round need.

Exposure by time window



Illustrative demo data

Mornings run worst in winter — the same routine needs different timing by season.

Jul ~95 · Satisfactory

Nov ~285 · Poor

Good
0–50

Satisfactory
51–100

Moderate
101–200

Poor
201–300

Very Poor
301–400

Severe
401+



§5 • HUMAN-CENTRED DESIGN

Personas & Daily Journey



Aarav, 20
STUDENT • WALKS TO CAMPUS

"Should I walk to campus today?"

PAIN

AQI numbers confuse; misses class or over-exerts.

NEED

Go / postpone verdict + safe-hour finder.

VAYUSATHI

2-line verdict + Hindi + commute tip.



Meera, 34
PARENT • TWO CHILDREN

"Can my child play outside?"

PAIN

Fear vs guesswork; no school guidance.

NEED

Child-safe windows aligned with school.

VAYUSATHI

Child mode + shareable school card.



Ramesh, 68
ELDERLY • ASTHMA-SENSITIVE

"Should I avoid going out?"

PAIN

Highest risk; generic alerts cause anxiety.

NEED

Calm, specific precautions.

VAYUSATHI

High-sensitivity profile + doctor line.



Salman, 29
DELIVERY RIDER • 8 HRS
OUTDOORS

"How can I work more safely?"

PAIN

Can't "stay indoors"; least informed.

NEED

Hindi harm-reduction that respects work.

VAYUSATHI

Mask / break / route tips. No shaming.



Kabir, 22
ATHLETE • CRICKET TRAINEE

"When should I train?"

PAIN

Fitness loss vs health-risk dilemma.

NEED

Best training slots + indoor swaps.

VAYUSATHI

48-hour practice planner.



City School
SCHOOL / CAMPUS • 1,200
STUDENTS

"Should outdoor sports shift?"

PAIN

No standard protocol for bad-air days.

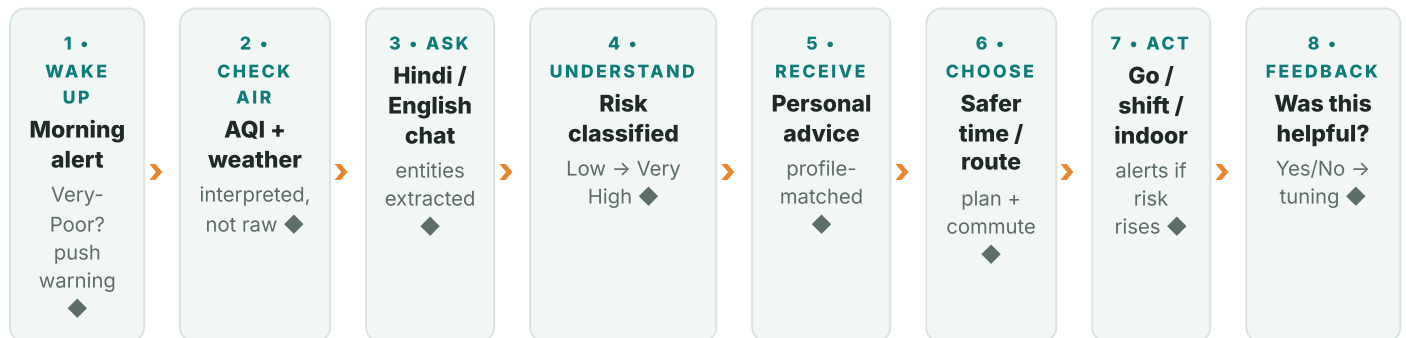
NEED

Rule + assembly announcement text.

VAYUSATHI

One-click printable morning card.

User journey — AI intervenes at every step ♦ = AI touchpoint





§6 • PURPOSE & POSITIONING

SDGs & Differentiation

PRIMARY • SDG 11

Sustainable Cities & Communities

Target 11.6: cut cities' per-capita environmental impact, incl. air quality. Safe-time planning + cleaner-commute nudges + campus monitoring culture.

SUPPORTING • SDG 3

Good Health & Well-being

Target 3.9: reduce illness from air pollution. Personal advisories for asthma, children, elderly; early warnings; school mode.

SUPPORTING • SDG 13

Climate Action

Target 13.3: climate education & awareness. Every advisory ends with a low-carbon tip — metro, carpool, no idling, no burning.

Why VayuSathi is different — generic comparison, no named competitors

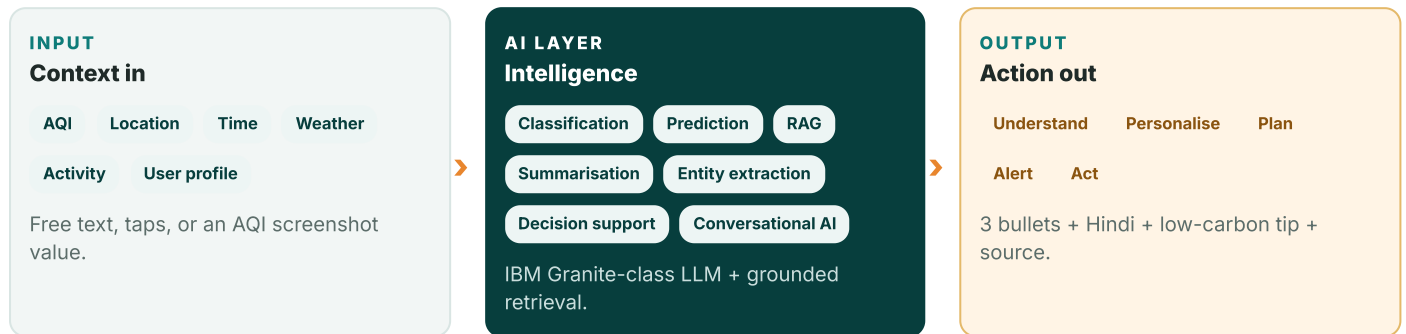
CAPABILITY	TYPICAL AQI APP	VAYUSATHI
AQI number	✓	✓
Plain-language explanation	Limited	✓ EN + Hindi, every answer
Personalisation by profile	Limited	✓ 6 profiles (student → athlete)
Hindi-first access	Limited	✓ summary + low-text cards
48-hour safe-window planning	Limited	✓ go / shift / indoor verdicts
Sustainable commute advice	Limited	✓ metro / carpool / walk-window
School / campus mode	—	✓ printable morning card + SOP
RAG-grounded, cited guidance	—	✓ CPCB / WHO / IMD sources cited
Responsible-AI guardrails	Limited	✓ fairness, transparency, safety, privacy



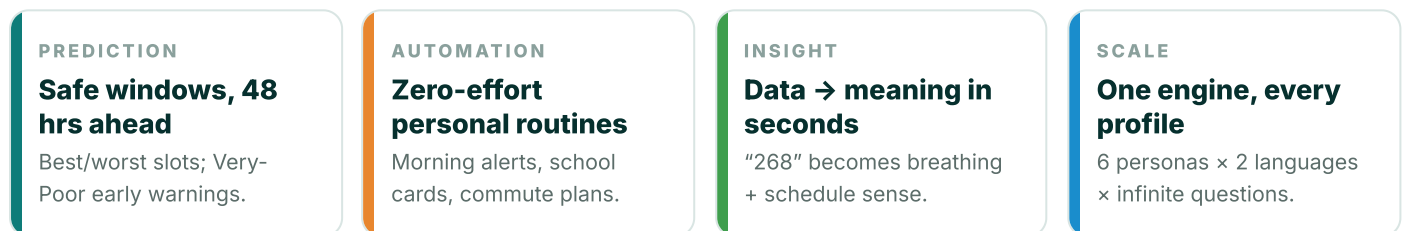
§7 • WHY AI IS NEEDED

AI Capability Map

Inputs change hourly — AQI, weather, user, language, activity — and the answer must be **personal**, **contextual** and **conversational**. Static charts and rule-based SMS cannot do this.



AI earns its place four ways



Allowed AI components in use (guideline §4)

Prompt engineering • IBM Granite-class models / watsonx-style orchestration • RAG • agentic workflow • entity extraction & summarisation • multimodal input • conversational UI. Logic and workflow design are the prototype — no advanced coding claimed.



§8 • MEET VAYUSATHI — FEATURES I

Six features, one assistant



F1 AQI Explainer

PURPOSE Turn any AQI value into plain EN + Hindi meaning.

INPUT → AQI + place + hour → band mapping + RAG summary → 3 do's/don'ts + Hindi + source.
AI → OUT

Example: "AQI 268 at Gomti Nagar, 8 AM" → *Poor — avoid 6–9 AM exertion; mask + low-traffic lane if out; windows shut till 10.*



F2 Personal Health Advisory

PURPOSE Risk-matched precautions for 6 sensitivity profiles.

INPUT → AQI + profile + duration → risk = f(band, profile, time) → HIGH/MODERATE verdict + precautions + doctor line.
AI → OUT

Example: Asthma student, 1 hr out → *HIGH: inhaler + split trip + mask + traffic-road avoid; breathless → see doctor.*



F3 48-Hour Outlook & Early Warning

PURPOSE Find best/worst windows; warn before Very-Poor spells.

INPUT → 48-hr slots + activity → window ranking + alert check → *go / shift / indoor* verdict with uncertainty stated.
AI → OUT

Example: Cricket slots → *Skip today; best Tue 2–5 PM (~140–150, Moderate). Forecast indicative.*

DEMO DATA



§8 • MEET VAYUSATHI — FEATURES II

Planning, campus & civic action



F4

Sustainable Commute Planner

PURPOSE Jointly minimise smoke exposure *and* emissions.

INPUT → Origin + destination + time + AQI → exposure × duration ranking → metro/carpool/walk plan + timing shift + low-carbon nudge.
AI → OUT

Example: Aliganj → University, 9 AM, AQI 240 → *Metro + 8-min masked walk; leave 9:40 to dodge the jam peak.*



F5

School & Campus Mode

PURPOSE A standard bad-air protocol for schools — zero hardware.

INPUT → Date + campus + morning AQI → rule table (≤100 normal ... 301+ suspend) → printable card + 40-word assembly text.
AI → OUT

Example: 18 Nov, AQI 285 → *INDOORS: PT inside; windows shut 8–10; asthma students alert teachers.*



F6

Civic Helper

PURPOSE Connect citizens, RWAs and the city on dust & burning.

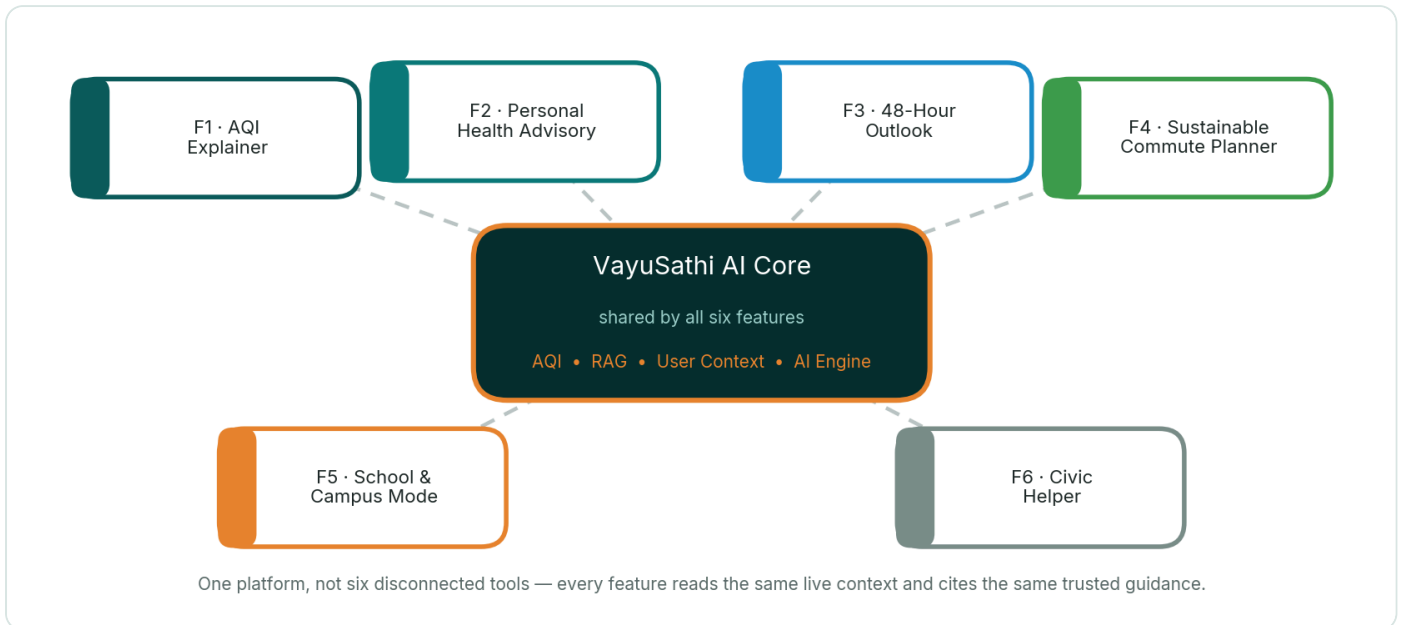
INPUT → Free-text complaints → ward-wise summarisation → weekly awareness post + routed complaint digest.
AI → OUT

Example: 40 mixed complaints → *Ward summary: 18 dust / 12 burning / 10 genset + draft notice-board post.*



§9 • ONE PLATFORM

Feature Ecosystem & Signature Pipeline



The six features share one AI core — same live context, same trusted guidance, same safety rules.

PROPOSED ARCHITECTURE

Data → Decision — the signature pipeline



Why this matters to an evaluator

Each stage is independently testable: data freshness is declared, intelligence is cited, actions are bounded by the prompt contract (§12), and impact is measured by the KPI framework (§19) — never claimed in advance.



§10 • TECHNICAL ARCHITECTURE

System Architecture — Eight Layers

Every user question flows top-to-bottom through eight testable layers, then back up through a feedback loop.

PROPOSED ARCHITECTURE

1

User inputs

Text (Hindi/English) • location • time • activity • AQI screenshot value • user profile

2

Input processing

Entity extraction (place + time + activity) • validation • context detection. *Invalid location → clarify.*

3

Data sources (RAG knowledge base)

CPCB bands • WHO guidance • IMD patterns • municipal norms • school rule table. *6–10 curated chunks.*

4

RAG — retrieve → rank → validate → cite

Top-3 chunks retrieved, band-checked, cited in the answer. *Stale (>3 hr) → declared, never hidden.*

5

AI engine

IBM Granite-class LLM with watsonx-style orchestration. *No fine-tuning claimed.*

6

Decision logic

Assess risk (Low → Very High) • personalise by profile • plan windows & routes • generate advisory.

7

Safety & Responsible AI

TRUST LAYER

Fairness • transparency • privacy • medical boundary • data freshness • fallbacks. *Full detail §17.*

8

Outputs

Chat answer • Hindi summary • alerts • commute plan • school card • 48-hour planner.

↻ Feedback loop — “Was this helpful? Yes / No”

Every answer ends with a vote. Votes + corrections feed a **weekly prompt review** (LEARN → MONITOR), closing the agentic loop (§11).



§11 • GROUNDING & AGENCY

RAG Pipeline & Decision Loop

How answers stay grounded

1 Trusted sources
CPCB • WHO • IMD • municipal • school rules

2 Chunking → knowledge base
Short, band-tagged guidance chunks

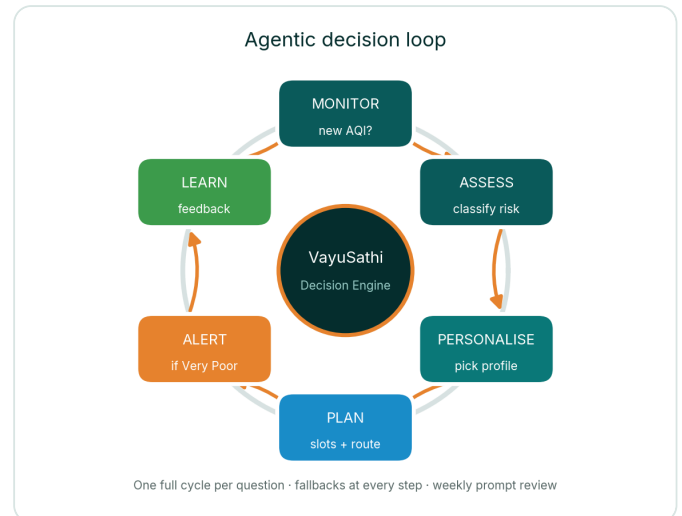
3 Retrieval + AQI validation
Top-3 chunks; band match checked

4 LLM generation + citation
Answer cites its source band

5 Safety filter → response
Contract check, then Hindi summary

Trust guarantee

"If data is missing or stale, VayuSathi does not invent an AQI value." It says so — and falls back to seasonal guidance.



Agentic loop: MONITOR → ASSESS → PERSONALISE → PLAN → ALERT → LEARN. **PROPOSED ARCHITECTURE**

Cycle discipline

One full cycle per question • a fallback at every step (§16)
• 1 LLM call + 1 retrieval per workflow — pilot-ready on low-cost tiers.



§12 • FOR TECHNICAL EVALUATORS

Under the Hood

THE AI EQUATION

LLM + prompt engineering + RAG + entity extraction + risk classification + agentic workflow + safety guardrails

= VayuSathi

Each term is visible in the workflows (§13) — nothing is a black box.

TECH STACK

AI	IBM Granite-class model • watsonx-style assistant
Knowledge	CPCB • WHO • IMD • municipal guidance
Methods	Prompt eng. • RAG • summarisation • entity extraction • decision support
Interface	Conversational UI • Hindi/English • mobile / web

DATA TRUST

CPCB

WHO

IMD

Municipal

UN SDGs

Retrieve



Verify



Generate



Cite

No fine-tuning claimed. Sources stay inspectable in the knowledge base.

PROMPT CONTRACT — EVERY ANSWER

- ✓ Max 3 action bullets
- ✓ 1 Hindi summary
- ✓ 1 low-carbon tip
- ✓ Source cited (band + origin)
- ✓ “Helpful? Yes/No” feedback
- ✓ Medical safety line when relevant



§13 • PROTOTYPE — WORKFLOW 1 OF 5

P1 • AQI Explainer

Purpose: turn any AQI value into plain EN + Hindi meaning. DEMO DATA

```
# SYSTEM PROMPT (excerpt)
Role: explain any AQI value in plain words.
Slots: {aqi} {place} {hour} {rag_band_guidance}
Steps: map band → 1-line breathing meaning → 3 do's/don'ts (RAG only)
→ 2-line Hindi summary → source line. No medical claims. Grade-6 language.
```

Inputs

AQI 268 • Gomti Nagar • 8 AM +
CPCB Poor-band chunk

Processing

Band mapping → RAG summary →
Hindi render → contract check

Output contract

3 bullets + Hindi + tip + source +
helpful?

USER

Explain AQI 268 at Gomti Nagar, 8 AM.

VAYUSATHI • 8:04 AM

268 **POOR**

What it means: heavy dust/smoke — a long morning walk can leave you breathless.

Recommendation: avoid 6–9 AM exertion → indoor walk, or after 4 PM if AQI < 200. **Safer option:** mask + low-traffic lane + water. **At home:** windows shut 7–10 AM.

Hindi: *Subah bahar lambi sair na karein. Ghar me walk karein. Bahar jayein to mask lagayein.*

Clean-air tip: metro / carpool today; no idling.

Source: CPCB Poor-band guidance • Was this helpful? Yes / No

Safety guardrail

No diagnosis • band + source cited • stale data would be declared, never gap-filled.



§13 • PROTOTYPE — WORKFLOW 2 OF 5

P2 • Personal Advisory

Purpose: risk-matched precautions for six sensitivity profiles — same risk logic, adapted guidance.

DEMO DATA

```
# SYSTEM PROMPT (excerpt)
Profiles: student | child | elderly | asthma | outdoor_worker | athlete
risk = f(band, profile, duration) → Low / Moderate / High / Very High
Steps: classify → profile precautions (≤3) → escalation line → Hindi tone-match.
Never diagnose. Never ask for medical history.
```

Inputs

Asthma student • AQI 268 •
Hazratganj • 1 hr outdoors

Processing

Poor × 1 hr × asthma → **HIGH** →
precaution template → safety line

Output contract

Risk verdict + 3 precautions + Hindi
+ doctor line + source

USER

I'm an asthma student. AQI 268, out for 1 hour — precautions?

VAYUSATHI

Risk for asthma profile: HIGH (Poor × 1-hr outdoor).

① Carry inhaler/medicine; avoid jam roads and exhaust side. ② Split the trip (30 min + indoor break); fitted mask; sip water. ③ If wheezing/breathless — stop, rest, **see a doctor promptly**.

Hindi: *Asthma walo ke liye aaj risk ZYADA. Dawa saath, mask, bheed wali sadak se bachein.*

General info only, not medical advice • CPCB Poor-band + WHO sensitive-group pattern • Helpful? Yes / No

Safety guardrail

Fairness: identical risk engine for all profiles — only precautions and language adapt. Doctor-escalation line is mandatory.



§13 • PROTOTYPE — WORKFLOW 3 OF 5

P3 • 48-Hour Outlook

Purpose: rank best/worst time windows and raise early warnings before Very-Poor spells.

DEMO DATA

SYSTEM PROMPT (excerpt)

Slots: {today_slots[6]} {tomorrow_slots[6]} {weather} {activity}

Steps: rank windows → verdict (go / shift / indoor-swap)

→ if any slot ≥301: EARLY WARNING + night precautions.

State uncertainty ("likely", "may change"). Forecast ≠ measured fact.

Inputs

Cricket • today 210–280 • tomorrow
140–200 improving

Processing

Window ranking → best slot → alert
check (max 280: no Very-Poor alert)

Output contract

Verdict + today/tomorrow plan +
watch line + Hindi + tip

USER

Cricket practice in Lucknow — today or tomorrow? When is best?

VAYUSATHI

Verdict: skip today morning → best window Tue 2–5 PM (likely Moderate ~140–150).

Today: Poor all day — indoor fitness; ground session ≤30 min with breaks. **Tomorrow:** improving after 12 PM — nets 2–5 PM, water, indoor warm-up. **Watch:** haze thickens or 300+ → fully indoors.

Hindi: Aaj maidan me lamba khel na karein. Kal 2–5 baje behtar sambhavna — wahi slot rakhein.

Forecast indicative, may change • Team carpool/metro tip • Helpful? Yes / No

Safety guardrail

Uncertainty is always stated; the full 48-hour trend visual lives on the city dashboard (§15).



§13 • PROTOTYPE — WORKFLOW 4 OF 5

P4 • Commute Planner

Purpose: jointly minimise smoke exposure *and* emissions — the SDG 11 × SDG 13 link in one answer.

DEMO DATA

```
# SYSTEM PROMPT (excerpt)
Slots: {origin} {destination} {time} {aqi_now} {options}
Steps: rank by (exposure × duration), then emissions → timing shift?
→ 1-line low-carbon nudge.
Walk/cycle on Poor+ days: mask + low-traffic route required, else recommend against.
```

Inputs

Aliganj → University • 9 AM • AQI 240 • metro / shared auto / bike

Processing

Exposure ranking → 9:40 shift check
→ nudge selection

Output contract

Ranked options + timing + active-travel rule + Hindi + tip

USER

Aliganj to University at 9 AM. AQI 240 — best option?

VAYUSATHI

Best today: Metro + 8-min walk (masked) — lowest smoke + lowest carbon.

Rank: ① Metro + short walk ② Shared auto / carpool ③ Solo two-wheeler — avoid (max exhaust). **Timing:** 9:40 if possible; dodge the 8–9 jam peak. **On foot/cycle:** inner lane, mask, steady pace, water break.

Hindi: *Aaj metro + thodi paidal (mask ke saath) sabse behtar. Jaam se bachein.*

CPCB Poor-band + exposure heuristic • No-idling tip • Helpful? Yes / No

Safety guardrail

Active-travel advice is conditional on band + mask + route — never a blanket “cycle more” on red days.



§13 • PROTOTYPE — WORKFLOW 5 OF 5

P5 • School Advisory Card

Purpose: a printable morning card + assembly text — a zero-hardware bad-air SOP for schools.

DEMO DATA

```
# SYSTEM PROMPT (excerpt) — rule table: ≤100 Normal • 101-200 Shortened • 201-300 Indoors • 301+ Suspended  
Slots: {date} {campus} {aqi_morning} {forecast}  
Steps: status → 40-word assembly text (EN + Hindi line) → card fields  
(date, AQI, colour, decision, 3 instructions).
```

Inputs

Univ. of Lucknow campus • 18 Nov •
AQI 285 • poor all day

Processing

285 → **INDOORS** → announcement
draft → instruction pick

Output contract

Card + assembly text + 3 instructions
+ source + helpful?

USER • SCHOOL ADMIN

Morning card for campus, 18 Nov. AQI 285, poor all day.

VAYUSATHI

CAMPUS ADVISORY • 18 Nov • AQI 285 (POOR) → Sports: INDOORS. PT inside; no morning jog on the main ground.

Assembly text: "Air quality is Poor today. Indoor games and hall practice only. If you go out, wear a mask and drink water. Hindi: Aaj hawa kharab hai — bahar khel-kood na karein, andar khelein."

Instructions: ① Windows shut 8–10 AM ② Asthma students keep medicine + inform teachers ③ Evening clubs → 3–4 PM, subject to update.

CPCB Poor-band + campus rule table • Demo — verify with CPCB reading • Helpful? Yes / No

Safety guardrail

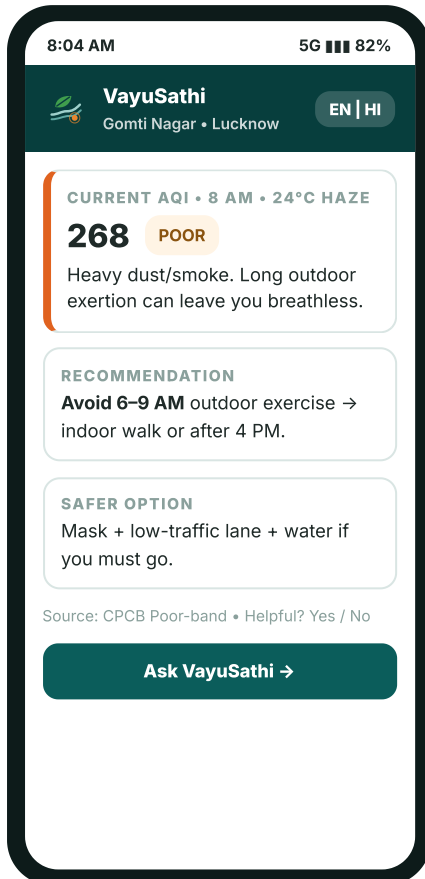
Human-in-the-loop: cards are marked "verify before assembly" — the school, not the model, makes the final call.



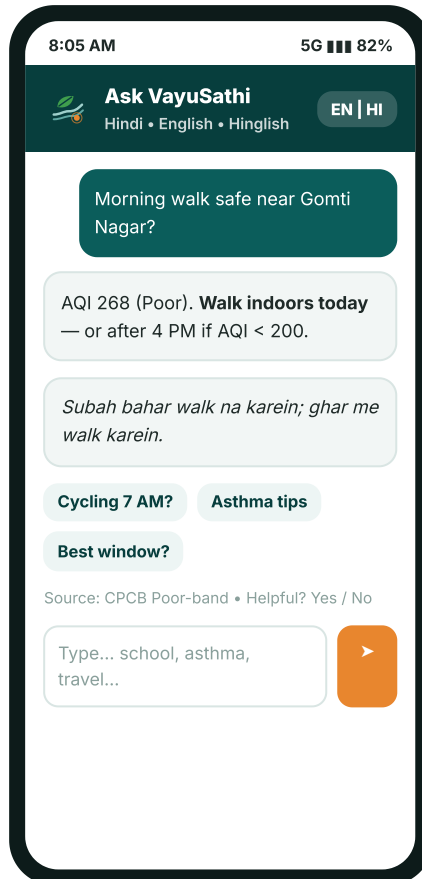
§14 • PRODUCT UI CONCEPT

Mobile Experience I

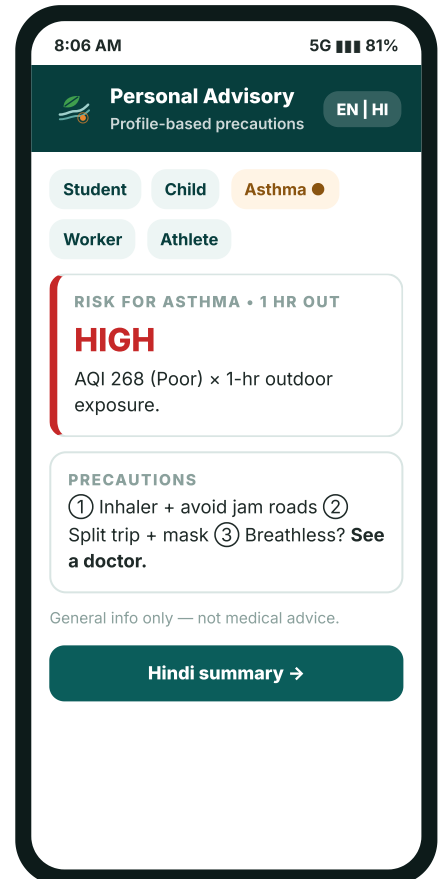
CONCEPTUAL UI



1 • Home — AQI + verdict in 10 seconds

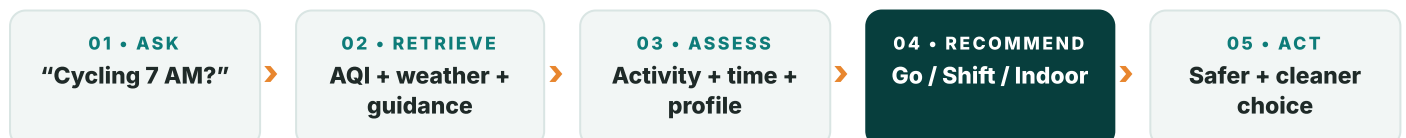


2 • Ask — chat in Hindi or English



3 • Personal — risk matched to you

The 30-second experience



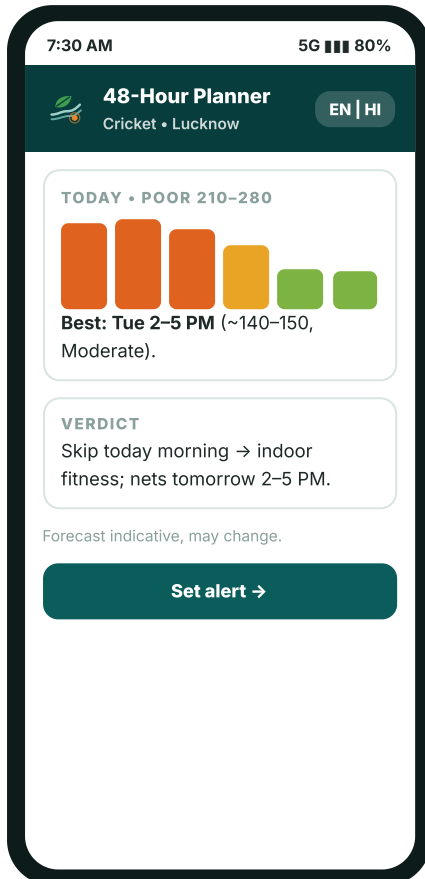


§14 • PRODUCT UI CONCEPT

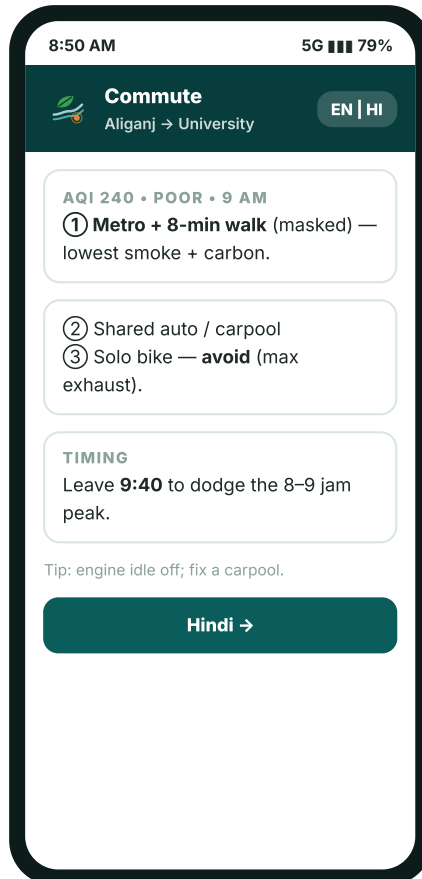
Mobile Experience II

CONCEPTUAL UI

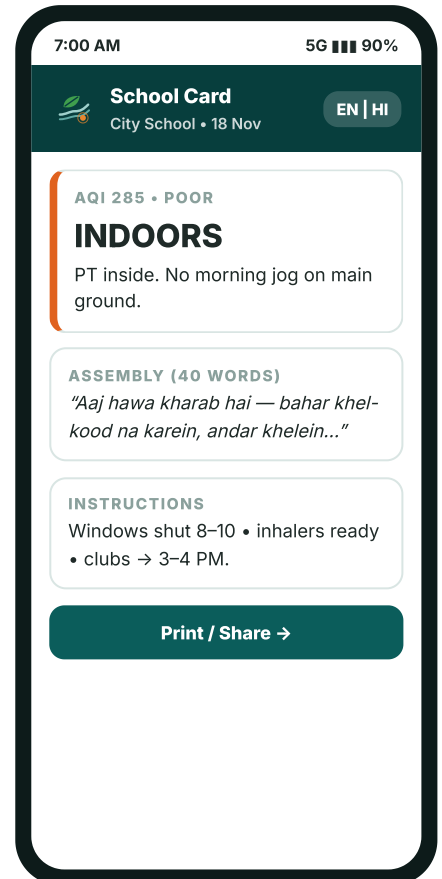
DEMO DATA



4 • Planner — best windows, 48 hrs



5 • Commute — cleaner + safer rank



6 • School — print-ready SOP card

Hindi mode — the same brain, your language

EN • ENGLISH

AQI 268 — Poor. Avoid 6-9 AM outdoor walk. Walk indoors, or after 4 PM if AQI drops below 200. Mask + water if you go out.

HI • HINDI — SCREEN 7

AQI 268 — Kharab. Subah bahar lambi sair na karein. Ghar me walk karein ya 4 baje ke baad jayein. Bahar jayein to mask + paani saath rakhein.

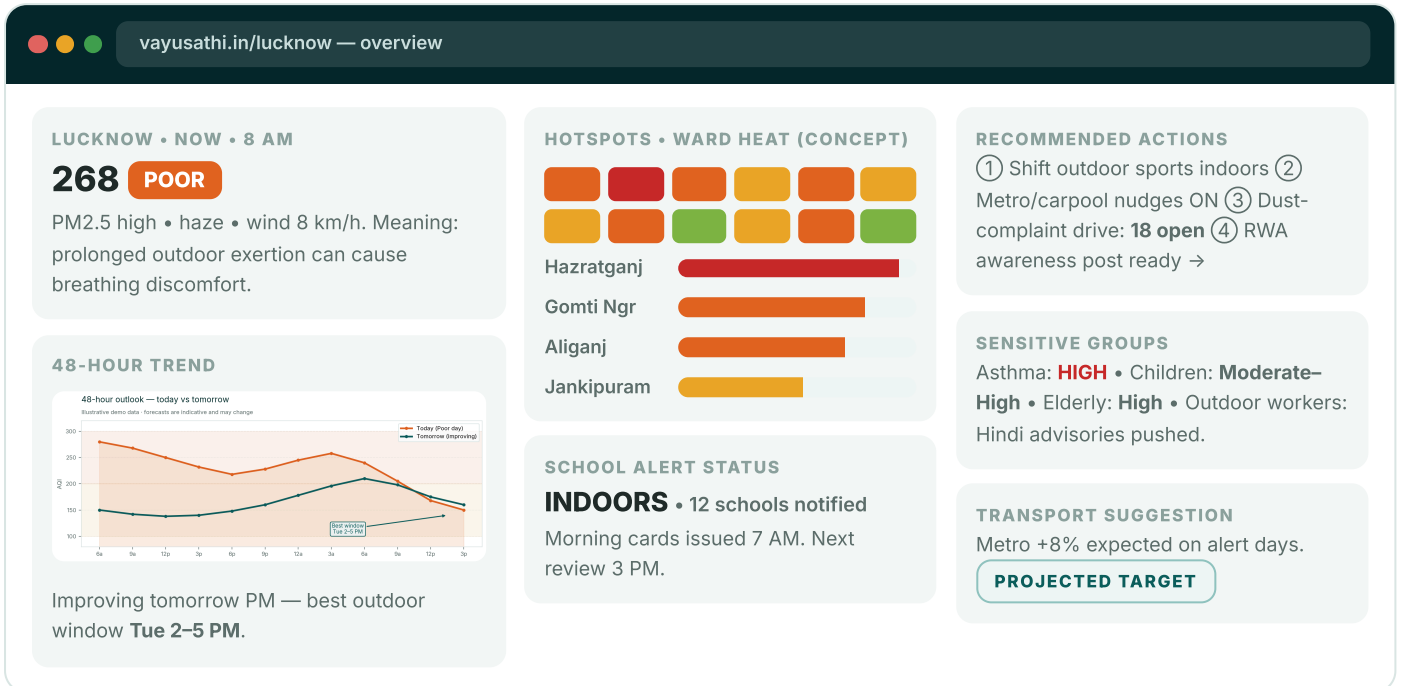


§15 • WEB CONCEPT FOR CAMPUSES & RWAS

City Dashboard Concept

CONCEPTUAL UI

DEMO DATA



Conceptual prototype — illustrative data. A production dashboard would bind to live CPCB feeds + IMD models (pilot scope, §19).



§16 • CREDIBILITY OVER HYPE

Live vs Prototype & Fallbacks

✓ EXISTS NOW

- ✓ 5 prompt workflows + contracts
- ✓ RAG design + knowledge plan
- ✓ 8-layer architecture
- ✓ Agentic loop logic
- ✓ Mobile + web UI mockups
- ✓ Sample outputs + test scenarios

○ CONCEPTUAL

FUTURE SCOPE

- ✓ Live AQI API binding
- ✓ Production forecasting
- ✓ WhatsApp + voice
- ✓ Clinical validation
- ✓ Deployment statistics

→ NEXT STEP

Campus pilot (4–6 weeks): live AQI feed, web chat, 1 campus + 1 school, Yes/No feedback loop, end-season survey. Full plan §19.

Nothing on the “conceptual” list is presented as done anywhere in this report.

Failure & fallback matrix — tested, not assumed

WHEN THIS HAPPENS	VAYUSATHI RESPONDS	WHY IT MATTERS
AQI missing	“Data unavailable” + seasonal guidance	Never invents a number
AQI stale (>3 hr)	Uncertainty banner + cautious advice	Freshness is declared
Invalid location	Asks user to clarify area	No silent wrong-city answers
Hindi / Hinglish query	Hindi-first response, same logic	Fairness across languages
Very-Poor forecast (300+)	Early warning + indoor default	Protection before exposure
Breathing difficulty mentioned	General info + “see a doctor promptly”	Hard medical boundary



§17 • MANDATORY — AND CENTRAL TO THE DESIGN

Responsible AI Trust Layer



RESPONSIBLE AI

enforced in every prompt (§13)

What VayuSathi will never do

- × No invented AQI values
- × No medical diagnosis
- × No hidden personal profiling
- × No false certainty
- × No blaming outdoor workers
- × No silent stale data

FAIRNESS

Same risk logic, all users

Only precautions & language adapt.
EN/HI gap monitored.

TRANSPARENCY

Every answer shows its basis

AQI + band + source cited; forecasts marked "likely".

SAFETY

No medical diagnosis

Breathing difficulty → doctor line, always.

PRIVACY

Minimal, session-only

No name/phone; area-level location;
no history asked.

DATA FRESHNESS

Stale data is declared

>3 hr → banner + seasonal fallback.

HUMAN OVERSIGHT

Verifiable decisions

School cards checked before assembly.

■ Safety flow — understandable in 10 seconds





\$18 • IF PILOTED — ONE WINTER, ~5,000 USERS

Impact Dashboard

PROJECTED TARGETS — NOT RESULTS

HEALTH

Breathe safer, act earlier

- -25-35% long outings on Poor+ days
- 80%+ recall their safe window
- Earlier care-seeking for breathlessness

ENVIRONMENT

Cleaner trips, less smoke

- +15-20% metro/carpool on alert days
- Less idling via timing nudges
- 100+ dust/burning complaints routed

SOCIAL

Access for everyone

- 40%+ users in Hindi mode
- 2 schools adopt the morning-card SOP
- Worker-friendly, zero-shame guidance

INSTITUTIONAL

A reusable playbook

- Campus SOP + RWA awareness kits
- Data-to-action culture for heat/rain/waste
- Helpfulness Yes-rate $\geq 75\%$

Impact logic model — how inputs become outcomes





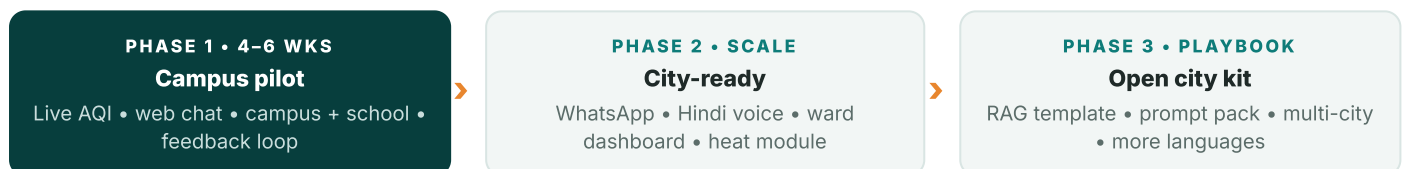
§19 • FROM PILOT TO PLAYBOOK

Measure, Roadmap & Scale

KPI framework PROJECTED PILOT TARGETS

METRIC	TARGET	MEASUREMENT METHOD
Long outdoor exposure on Poor+ days	-25-35%	In-chat poll + end-season survey
Safe-window recall	≥ 80%	3-question recall survey
Metro / carpool on alert days	+15-20%	Commute-mode poll on alert days
Hindi-mode usage	≥ 40%	Session language analytics
School SOP adoption	2 schools	Signed morning-card logbook
Helpfulness Yes-rate	≥ 75%	"Was this helpful?" votes

Pilot roadmap



Scalability — Lucknow today, multi-hazard platform tomorrow FUTURE SCOPE

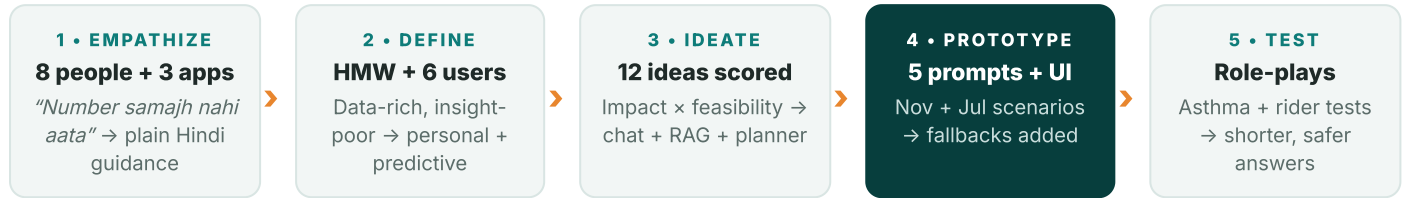




§20 • HOW THE SOLUTION WAS FOUND

Design Process & Accessibility

Design thinking — action → insight → decision



Traceability — from pain to impact

USER PAIN	INSIGHT	FEATURE	AI CAPABILITY	IMPACT
AQI confusion	Need simple guidance	F1 Explainer	RAG + summarisation	Faster decisions
Generic warnings	Risk differs by person	F2 Advisory	Classification	Protected sensitive groups
Guessed timings	Need safe windows	F3 Outlook	Prediction	Shifted exposure hours
No school protocol	Need a standard rule	F5 Campus mode	Decision support	Safer school routines

Accessibility — an inclusive-design strategy

Hindi / English Large text Short sentences Voice-ready Low-text cards Colour + text AQI (never colour alone) WhatsApp-friendly forwards

Every choice serves a persona: low-text cards for riders, voice-ready lines for elderly users, colour-plus-text AQI for colour-blind users, WhatsApp formats for feature-phone households.



§ 21 • ANNEXURE

Transparency & Sources

CPCB

AQI bands, breakpoints & band precautions (paraphrased). Basis of all demo logic.

WHO

Global air-quality guideline values; sensitive-group patterns.

IMD

Lucknow seasonal & weather patterns (inversion, dust phases).

Municipal guidance

Dust-control & waste-burning advisory patterns; complaint categories.

UN SDGs

Targets 11.6, 3.9, 13.3 mapped in §6.

Internship guideline

1M1B project-creation doc §§1–9; §8 deliverables mapped below.

Methodology notes

Conceptual prototype: 5 prompt workflows with I/O contracts, 8-layer architecture, agentic loop, 9 UI screens, 8 custom visuals. Tested on Nov Poor-day + July Satisfactory scenarios and 6 failure cases (§16). **AI approach:** prompt engineering + RAG + agentic workflow on IBM Granite-class models — **no fine-tuning claimed.**

Demo-data disclaimer

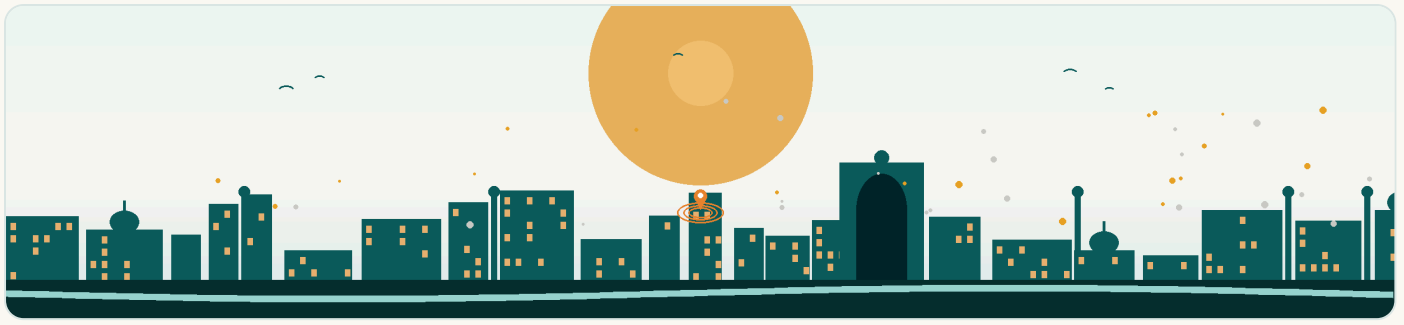
Every AQI number, forecast slot and chart on pp. 4–6, 14–16, 21–23 is **illustrative demo data** inspired by typical Lucknow seasonal patterns — labelled **DEMO DATA** wherever shown. Not live monitoring; not measurements.

Guideline §8 map

Description → §§1–7, 12, 17–18 • **Prototype/demo** → §§9–15 (prompts + diagrams + UI) • **Impact statement** → §18. Responsible-AI §7 → §17. Design thinking §5 → §20.

What was NOT fabricated

No real-time readings, adoption, clinical, accuracy, API-integration or deployment claims appear anywhere. Targets are **PROJECTED TARGET**; UI is **CONCEPTUAL UI**; architecture is **PROPOSED ARCHITECTURE**.



CLOSING • VAYUSATHI

"From AQI numbers to everyday action."

VayuSathi — Helping Lucknow Breathe Better. A thoughtfully designed AI companion: personal, predictive, Hindi-first, and safe by construction.

PROBLEM

Air data is hard to act on

AI SOLUTION

LLM + RAG + planning

USERS

Students • families • workers • athletes • schools

OUTPUTS

Advice • alerts • windows • commute plans

SUSTAINABILITY

Healthier habits + cleaner mobility

STATUS → NEXT

Conceptual prototype → campus pilot

Mandeep Singh

University of Lucknow • Lucknow, India



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